

Harper Lee

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About Me

Hardworking Lead programmer with 4+ years of experience in game development and software engineering, specializing in creating interactive experiences using Arduino and Raspberry Pi. Proven track record in competitive game jams, consistently ranking in the top 20%. Skilled in writing maintainable code and documenting projects effectively. Experienced in managing open-source projects and developing scalable systems, with a strong foundation in various programming languages and design patterns.

Technical Skills

- Programming Languages: Python, Lua, C#, Rust, Go, Java, TypeScript
- Hardware: Arduino, Raspberry Pi, Automation, Lighting, Electrical, Soldering
- Software: Vim, VSCode, Zed, Adobe Suite, Microsoft Office, Unity, Godot, Node-Red
- Operating Systems: Linux, Windows, MacOS
- Virtual Machines: Proxmox, VMWare, VirtualBox, VirtKVM
- Productivity: Jira, Trello, Notion, ClickUp, Git, GitHub

Experience

Beat The Lock Escape Rooms

Game Developer

February 2025 – Ongoing

Santa Clara, California

- Developed custom game-host software to manage puzzle logic, room states, and real-time player feedback.
- Engineered and deployed custom Linux services on Raspberry Pi to control and automate physical escape room puzzles.
- Designed and implemented a web-based room and puzzle management interface using Node-RED, enabling real-time monitoring and manual overrides.
- Integrated MQTT to facilitate reliable, low-latency communication between puzzle components and a central control system.
- Authored comprehensive design and technical documentation for new puzzles, detailing hardware schematics, software architecture, and interaction design.
- Redesigned legacy puzzles to modernize user inputs, improve reliability, and enhance the player experience by introducing intuitive, responsive interaction methods.

Key Skills:

- **Technologies:** Raspberry Pi, Linux system services, Node-RED, MQTT
- **Programming:** Python, JavaScript, embedded systems
- **Design:** Puzzle design, UX for physical interactions, rapid prototyping
- **Other:** Technical documentation, systems integration

OMESCAPE - Real Room Escape Game

Lead Programmer

January 2021 - August 2024

Sunnyvale, California

- Developed maintainable, scalable software enabling system interaction via GPIO and WebSockets.
- Engineered interactive experiences using Arduino and Raspberry Pi, blending physical and digital elements.
- Authored clear, comprehensive project documentation tailored for both technical maintainers and non-technical game masters.
 - Designed for easy understanding by less technical readers.
 - Included detailed architecture and maintenance guidance for developers.

- Built and administered Linux-based servers and databases to support puzzle logic, logging, and automation.
- Implemented a local VoIP system for in-room communication between players and hosts.
 - Integrated with rotary phones to preserve thematic immersion.
 - Included playback of context-specific pre-recorded messages.
- Delivered projects with high customer satisfaction and minimal need for post-deployment adjustments.
- Operated independently on end-to-end development, from concept to deployment and support.

Key Skills:

- **Technologies:** Embedded systems, Arduino, Raspberry Pi, GPIO, WebSockets, VoIP (Asterisk or similar)
- **Programming:** Python, C/C++, shell scripting
- **Systems:** Linux server administration, hardware-software integration
- **Documentation:** Technical writing, user documentation

Private Projects

- **Cicada:** An easy to use python library for creating Escape Room puzzles using Raspberry Pi and Raspberry Pi Pico, built to feel like a Game Engine.
- Self-hosted Proxmox cluster running Ubuntu Server containers for open-source software.
 - Runs virtualized Windows 10 with PCIe Passthrough and low-latency remote viewing.
 - Runs several common home server softwares, including Plex, NextCloud, HomeAssist, and others.

Education & Training

- **Undergraduate Coursework,** Game and Interactive Media Design; Academy of Art University 2020 - 2023
- **Unity Programming Certification;** Unity Certification Number: 90088844

Interests

Competitive Game Jams Participant

- Participated in 6 public game jams as lead developer and director.
- **GitHub Game Off:** Scored 8th in Gameplay, 23rd Overall.
- Other events include Ludum Dare and GMTK Game Jam, consistently scoring in the top 20%.